

# Chitragupta

Production Deployment Guide — company ERP go-live

## Topology

Browser (ERP Desk + widget) → ERPNext (HTTPS) and → Chitragupta backend (HTTPS, separate host or same server). Backend → ERPNext REST as a narrow bot user (fallback) or as the real person (bound session).  
Backend → Anthropic API. Database: PostgreSQL.

## Step 1 — The bot user (keep it narrow)

- Create chitragupta-bot@company.com with ONLY the roles for live modules (start: Purchase User). Never Administrator in production.
- User → API Access → Generate Keys.
- Verify with check\_connection.py — 403s on payroll/GL are the fence working.

## Step 2 — Backend as a service (Linux + systemd)

```
python -m venv /opt/chitragupta/venv && pip install -r requirements.txt
/etc/chitragupta.env (chmod 600): ERP_URL, ERP_API_KEY, ERP_API_SECRET, ANTHROPIC_API_KEY,
CHITRAGUPTA_SSO_SECRET (openssl rand -hex 32), CHITRAGUPTA_DB=postgresql+psycopg2://...
systemd unit: uvicorn app:app --host 127.0.0.1 --port 8001, Restart=always
nginx in front with TLS. Verify https://chitragupta.company.com/api/version
```

## Step 3 — The Desk widget (Frappe app)

- bench new-app chitragupta\_desk; widget JS → public/js/ with line 1 window.CHITRAGUPTA\_URL = "https://chitragupta.company.com";
- chitragupta\_desk\_api.py → chitragupta\_desk/api.py (the SSO method)
- hooks.py: app\_include\_js = ["/assets/chitragupta\_desk/js/chitragupta\_desk.js"]
- bench --site SITE install-app chitragupta\_desk; bench build --app chitragupta\_desk (bench RUNNING — redis warnings mean the asset didn't register); clear-cache
- Browser: Empty Cache and Hard Reload. Widget updates always need: re-pull JS → build → clear-cache → hard reload.

## Step 4 — Act-as-user (SSO)

- bench --site SITE set-config chitragupta\_sso\_secret ""; bench restart
- Verify /api/method/chitragupta\_desk.api.get\_session\_assertion returns signed JSON
- Widget header shows the REAL name + roles; backend log shows POST /api/sso 200
- CLOCKS MUST AGREE: NTP on both hosts; on VMs disable the hypervisor's competing time sync. Replay window 900s (CHITRAGUPTA\_SSO\_MAX\_AGE).

## Go-live checklist

| # | Check                                                                  |
|---|------------------------------------------------------------------------|
| 1 | /api/version shows the deployed build; test suite passes on the server |
| 2 | check_connection.py: bot correctly fenced (403 on payroll/GL)          |

| # | Check                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------|
| 3 | SSO end-to-end: real roles in header; forged/stale assertions rejected 401                            |
| 4 | Draft→approve→read-back on a test PO; owner = the real user                                           |
| 5 | HTTPS everywhere; secrets only in env file; ALL keys rotated (dev keys appeared in chats/screenshots) |
| 6 | PostgreSQL live; nightly backup job scheduled and restore tested                                      |
| 7 | Limited-user test: a Purchase-only account is correctly scoped; Sales Invoice refused for them        |
| 8 | /api/learning responding; starting clean_rate noted for the weekly chart                              |

## Rollout plan

- Week 1–2: 20-user Purchase pilot, drafts-only; measure adoption + clean-approval rate weekly
- Expand roles module-by-module (add roles to the bot; ERPNext scopes each real user automatically)
- Only after adoption is proven: WhatsApp channel, multilingual, automode